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(An Autonomous Institute Affiliated to Savitribai Phule Pune University)

Name - Prachi Ganesh Lohkare

PRN - 202401040025

Batch - A1

Roll No. - 04

AIML IA ACTIVITY - 1

Report on AI Applications using PEAS Analysis, Intelligent Agents, and Environment Types

Application 1: AI-Based Smart Mirror

1. Working -

- Gives skincare analysis.
- Detects facial fatigue.
- Suggests makeup or grooming tips.

2. General PEAS Analysis -

PEAS Component	Detailed Description
Performance Measure	Accurate facial/skincare analysis, useful recommendations, quick response, and user satisfaction.
Environment	Home, bathroom, or salon where the user interacts with the smart mirror.

Actuators	Display, speakers, visual/audio notifications, and recommendation interface.
Sensors	Camera, microphone, proximity sensor, user feedback, and stored interaction/history data

3. 10 Agent–Environment Combinations with Individual PEAS

Combination 1: Simple Reflex Agent + Fully Observable Environment

Basic information display when the required state is completely available.

PEAS Component	Detailed Description
Performance Measure	Correct identification of the user's presence, accurate information display, and fast response.
Environment	Home, bathroom, or salon where the user is clearly visible; relevant information is available. Fully observable for the particular task.
Actuators	Display, speakers, and visual notifications.
Sensors	Camera, proximity sensor, and microphone.

I observed that a simple reflex agent can directly connect a sensor condition with an action when the information needed for the decision is available.

Example: User stands in front of the mirror → proximity sensor detects the user → mirror displays weather or time.

Combination 2: Simple Reflex Agent + Static Environment

Basic functions such as displaying time or predefined information.

PEAS Component	Detailed Description
Performance Measure	Accurate display of time/date or other basic information and reliable response.
Environment	A fixed mirror location where the relevant task does not significantly change while the action is performed. Static for this task.
Actuators	Display and speakers.
Sensors	Proximity sensor, motion/touch input, and time input.

I observed that this is suitable for simple mirror functions because the agent follows a direct rule without needing complex reasoning or history.

Example: Mirror is activated → system detects activation → current time and date are displayed.

Combination 3: Model-Based Agent + Partially Observable Environment

Using previous facial observations when the current facial information is incomplete.

PEAS Component	Detailed Description
Performance Measure	Accurate facial analysis, improved recommendations, and consistent interpretation of incomplete information.
Environment	Home, bathroom, or salon where lighting, camera angle, and visible facial information can vary. Partially observable.

Actuators	Display, speakers, and notifications.
Sensors	Camera, microphone, proximity sensor, and previous facial observations/history.

I observed that one facial image cannot reveal everything about the user's condition. A model-based agent can use previous observations as part of its internal state.

Example: Today's facial image + previous observations → comparison → improved skincare advice.

Combination 4: Model-Based Agent + Sequential Environment

Tracking changes in facial or skin condition over multiple interactions.

PEAS Component	Detailed Description
Performance Measure	Correct tracking of changes, meaningful comparisons, and useful long-term recommendations.
Environment	User's home, bathroom, or salon where observations occur over time and previous states are relevant. Sequential.
Actuators	Display, speakers, and notifications.
Sensors	Camera, microphone, proximity sensor, and historical facial/skin data.

I observed that skincare monitoring is not a one-time task. Earlier observations can influence how later observations are interpreted.

Example: Day 1 observation → Day 2 observation → comparison → updated recommendation.

Combination 5: Goal-Based Agent + Dynamic Environment

Selecting actions to achieve a skincare-related goal while the user's condition changes.

PEAS Component	Detailed Description
Performance Measure	Achievement of the user's skincare goal, useful recommendations, and user satisfaction.
Environment	Home, bathroom, or salon where skin condition, facial appearance, lighting, and surroundings can change. Dynamic.
Actuators	Display, speakers, recommendation interface, and alerts.
Sensors	Camera, microphone, proximity sensor, and user input.

I observed that a goal-based agent is useful when the mirror has an objective rather than simply reacting to an input.

Example: Goal = improve skincare → camera analyses skin → AI identifies condition → suitable routine is suggested.

Combination 6: Goal-Based Agent + Continuous Environment

Continuously monitoring facial expressions to support a monitoring goal.

PEAS Component	Detailed Description

Performance Measure	Accurate continuous monitoring, correct fatigue detection, and timely alerts.
Environment	The user's facial expression changes continuously while interacting with the mirror. Continuous.
Actuators	Display, speakers, and visual/audio alerts.
Sensors	Camera, microphone, and proximity sensor.

I observed that facial expressions can change from normal to tired or sleepy within a short time, so continuous observation is useful.

Example: Camera continuously observes the face → signs of fatigue are detected → mirror provides a rest reminder.

Combination 7: Utility-Based Agent + Stochastic Environment

Selecting a useful skincare/product recommendation when outcomes are uncertain.

PEAS Component	Detailed Description
Performance Measure	Quality and suitability of recommendations, user satisfaction, and reduced unsuitable recommendations.

Environment	User's skin condition and possible product choices, where different users may respond differently. Stochastic.
Actuators	Display, speakers, and product recommendation interface.
Sensors	Camera, user preferences/input, and previous recommendation feedback.

I observed that the same skincare product may not give the same result to every user. A utility-based agent can compare several factors before recommending an option.

Example: AI evaluates several products → considers suitability and preference → recommends the option with higher expected usefulness.

Combination 8: Utility-Based Agent + Dynamic Environment

Balancing cost, quality, suitability, and changing user requirements.

PEAS Component	Detailed Description
Performance Measure	Good balance between cost and quality, suitability, adaptability, and user satisfaction.
Environment	Changing skincare needs, product choices, prices, availability, and user preferences. Dynamic.
Actuators	Display, speakers, and recommendation interface.

Sensors	Camera, user preferences, product information, and previous feedback.

I observed that the most expensive product is not automatically the best choice. The system should balance several competing factors.

Example: Product A is expensive/high quality while Product B is affordable/suitable → AI compares utility → selects the more suitable option.

Combination 9: Learning Agent + Continuous Environment

Learning user preferences from repeated interactions.

PEAS Component	**Detailed Description**
Performance Measure	Improved personalization, better prediction of preferences, and increased user satisfaction.
Environment	Continuous interaction between the user and Smart Mirror, with preferences developing over repeated use. Continuous.
Actuators	Display, speakers, and personalized recommendations.
Sensors	Camera, microphone, proximity sensor, interaction history, and user feedback.

I observed that a learning agent can become more personalized because it can learn from repeated choices and feedback.

Example: User repeatedly selects simple skincare → AI learns the preference → future recommendations become more personalized.

Combination 10: Learning Agent + Partially Observable Environment
Improving advice using previous interaction and feedback when complete user information is unavailable.

PEAS Component	Detailed Description
Performance Measure	Improved recommendations despite incomplete information, better personalization, and fewer unsuitable suggestions.
Environment	Home, bathroom, or salon where complete facial, medical, or skincare information may not be directly observable. Partially observable.
Actuators	Display, speakers, and personalized alerts.
Sensors	Camera, microphone, proximity sensor, historical data, and user feedback.

I observed that the Smart Mirror cannot directly observe everything about a user. Learning from repeated interactions can help improve future advice despite incomplete information.

Example: Incomplete facial information + previous feedback → AI learns patterns → future advice is improved.

4. **Social Impacts**

- Promotes self-care.
- Saves time.
- Provides a personalized experience.
- Supports smart-home applications.

5. **Ethical Issues**

- Privacy of facial data.
- Biometric security.
- Algorithmic bias.
- User consent.

Application 2: AI-Based Personal Fitness Coach

1. Working

- Monitors workouts using a camera or smartwatch.
- Corrects exercise posture.
- Creates personalized workout plans.
- Tracks calories and heart rate.
- Adapts plans based on progress.

2. General PEAS Analysis

PEAS Component	Detailed Description
Performance Measure	Correct posture, accurate workout tracking, fitness improvement, useful personalization, and safe adaptation.
Environment	Gym, home, or park where the user performs physical activity.
Actuators	Mobile application, smartwatch, audio/visual alerts, notifications, and workout-plan interface.

Sensors	Camera, heart-rate sensor, accelerometer, motion data, workout history, and user feedback.

3. 10 Agent–Environment Combinations with Individual PEAS

Combination 1: Simple Reflex Agent + Fully Observable Environment

Counting repetitions when the complete movement required for the task is visible.

PEAS Component	Detailed Description
Performance Measure	Correct repetition counting, accurate exercise detection, and fast feedback.
Environment	Gym, home, or park where the user's complete relevant movement is visible. Fully observable for the particular task.
Actuators	Mobile application, smartwatch, and audio/visual feedback.
Sensors	Camera, accelerometer, and motion sensors.

I observed that when the complete movement is visible, the agent can directly connect an observed exercise state with a response.

Example: Camera detects completed squat → repetition count increases by one.

Combination 2: Simple Reflex Agent + Continuous Environment

Continuously monitoring posture during exercise.

PEAS Component	Detailed Description
Performance Measure	Accurate posture detection, immediate correction, and continuous monitoring.
Environment	User is exercising and body position changes continuously. Continuous.
Actuators	Mobile app, smartwatch, and voice/audio alerts.
Sensors	Camera, accelerometer, and motion sensors.

I observed that checking only the start and end of an exercise may miss incorrect movement in between, so continuous monitoring is useful.

Example: Incorrect posture detected during a squat → app immediately provides correction.

Combination 3: Model-Based Agent + Partially Observable Environment
Combining incomplete current sensor information with workout history.

PEAS Component	Detailed Description
Performance Measure	Accurate estimation of workout condition and appropriate recommendations.
Environment	Gym, home, or park where the camera may not capture the entire body and a

	smartwatch provides only some measurements. Partially observable.
Actuators	Mobile app, smartwatch, and audio/visual alerts.
Sensors	Camera, heart-rate sensor, accelerometer, and workout history.

I observed that one sensor may not provide complete information. Previous workout information can help the agent estimate the current state.

Example: Incomplete camera view + workout history + sensor data → AI estimates the exercise state.

Combination 4: Model-Based Agent + Sequential Environment

Adjusting future workout plans based on previous sessions.

PEAS Component	Detailed Description
Performance Measure	Improved workout planning, correct progress tracking, and suitable future adjustments.
Environment	Gym, home, or park where multiple workout sessions occur and previous performance affects later plans. Sequential.
Actuators	Mobile app, smartwatch, and workout-plan interface.

Sensors	Camera, heart-rate sensor, accelerometer, and workout history.

I observed that fitness improvement happens across multiple sessions, so previous performance is important when creating the next plan.

Example: Monday workout → performance recorded → next workout adjusted according to previous result.

Combination 5: Goal-Based Agent + Dynamic Environment

Creating actions and plans according to a specific fitness goal while conditions change.

PEAS Component	Detailed Description
Performance Measure	Achievement of fitness goals, fitness improvement, appropriate exercise selection, and user satisfaction.
Environment	Gym, home, or park where energy, performance, available time, and conditions can change. Dynamic.
Actuators	Mobile app, smartwatch, workout recommendations, and notifications.
Sensors	Camera, heart-rate sensor, accelerometer, and user input.

I observed that users have different fitness goals, so a goal-based agent can select exercises according to the desired outcome.

Example: Goal = improve endurance → AI recommends an endurance-focused workout.

Combination 6: Goal-Based Agent + Continuous Environment

Tracking heart rate and workout intensity continuously against a target.

PEAS Component	Detailed Description
Performance Measure	Accurate heart-rate monitoring, appropriate intensity, safe progression, and progress toward the workout goal.
Environment	User is continuously exercising and heart rate/movement change over time. Continuous.
Actuators	Mobile app, smartwatch, voice alerts, and notifications.
Sensors	Heart-rate sensor, accelerometer, and camera.

I observed that heart rate changes continuously during exercise, so the agent can compare the current state with the desired workout intensity.

Example: Target intensity → heart rate monitored continuously → AI recommends maintaining, increasing, or reducing intensity.

Combination 7: Utility-Based Agent + Stochastic Environment

Choosing a workout that balances expected fitness benefit and safety under uncertainty.

PEAS Component	Detailed Description
Performance Measure	Safe workout selection, fitness benefit, user comfort, and reduced unnecessary risk.
Environment	User's physical condition and response to different exercises are not perfectly predictable. Stochastic.
Actuators	Mobile app, smartwatch, audio alerts, and workout recommendations.
Sensors	Heart-rate sensor, camera, accelerometer, and previous workout data.

I observed that people can respond differently to the same exercise. Therefore, the system should balance benefit and safety rather than always choosing the hardest option.

Example: Two exercises provide similar benefit → AI uses available user data → chooses the safer option.

Combination 8: Utility-Based Agent + Dynamic Environment

Balancing workout effectiveness, safety, and changing intensity.

PEAS Component	Detailed Description

Performance Measure	Proper intensity management, fitness improvement, user comfort, and safe adaptation.
Environment	Gym, home, or park where the user's energy, heart rate, and performance change during the session. Dynamic.
Actuators	Mobile app, smartwatch, voice alerts, and exercise recommendations.
Sensors	Heart-rate sensor, camera, and accelerometer.

I observed that a fixed intensity may not remain appropriate throughout a workout. The agent should adapt while balancing effectiveness and safety.

Example: Heart rate becomes high → AI detects the change → recommends reducing workout intensity.

Combination 9: Learning Agent + Continuous Environment

Learning workout habits and preferences from repeated sessions.

PEAS Component	Detailed Description
Performance Measure	Better personalization, improved workout recommendations, accurate habit prediction, and long-term fitness improvement.
Environment	User performs repeated workouts and habits develop over time. Continuous.

Actuators	Mobile app, smartwatch, personalized workout plans, and notifications.
Sensors	Camera, heart-rate sensor, accelerometer, workout history, and user feedback.

I observed that a learning agent can identify patterns such as preferred workout times and frequently performed exercises.

Example: User regularly exercises in the morning → AI learns the pattern → future workouts are recommended around that routine.

Combination 10: Learning Agent + Partially Observable Environment

Improving recommendations from incomplete health and workout information.

PEAS Component	Detailed Description
Performance Measure	Better recommendations from available data, improved personalization, and reduced unsuitable recommendations.
Environment	Gym, home, or park where complete health information is not directly observable and sensor data may be incomplete. Partially observable.

Actuators	Mobile application, smartwatch, notifications, and workout recommendations.
Sensors	Heart-rate sensor, camera, accelerometer, workout history, and user feedback.

I observed that a fitness coach cannot measure every aspect of a person's health. It can learn from the information available, but incomplete data limits what it can safely conclude.

Example: Workout data + user response → AI learns from previous sessions → future recommendations are adjusted.

4. **Social Impacts**

- Encourages a healthy lifestyle.
- Provides more affordable coaching support.
- Improves fitness awareness.
- Supports home workouts.

5. **Ethical Issues**

- Health data privacy.
- User consent.
- Bias in recommendations.
- The system should not replace professional medical advice.